

Predicting Blendshapes of Virtual Humans for Low-Delay Remote Rendering using LSTM

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ABSTRACT

This paper proposes a novel framework to reduce the perceptual delay in rendering the facial expressions of virtual humans. The proposed method reduces the delay by predicting the future blend coefficients that represent facial expressions. The prediction accuracy of the proposed method is improved by 27% on average over a conventional LSTM. We also subjectively confirmed that the proposed method achieves natural facial expressions.

Index Terms: Computer systems organization—Real-time systems—Real-time system architecture Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models

1 INTRODUCTION

As the COVID-19 pandemic has limited face-to-face communication in the real world, virtual humans that embody users are expected to communicate in the virtual world. In particular, virtual humans that create a sense of presence ensure privacy and promote more subjective engagement in teleconferences [4]. However, the perceptual delay caused by processing and transmission is an issue for interactive communication. In this paper, we propose a novel framework to solve the problem of the delay by facial expression prediction. The proposed method is capable of predicting facial expressions considering temporal fluctuations in facial expression recognition.

2 RELATED WORK

If the user's motion can be predicted and applied to virtual humans, this will facilitate smooth communication by reducing the delay. Several methods have been developed to predict the motion of virtual humans. Gül [3] uses a Kalman filter to predict head movements 60 msec ahead to reduce the delay in server-side rendering. Lighterbody [6] uses QuaterNet [8] to predict body movements 100 msec in the future based on the rotation angle of the skeleton estimated by Kinect V2. However, neither method can handle detailed movements such as facial expressions. Audio2face [9] predicts blendshapes that represent facial expressions from audio using a long short-term memory network (LSTM) [5]. However, audio2face is unable to reproduce facial expressions such as smiling when the user is not speaking.

Blendshape is a technique for creating multiple deformable shapes in advance and animating them by moving the vertices of the original shapes. Blendshape generates an arbitrary expression S' by weighting the difference between the base shape S_0 and the k -th expression models S_k with blend coefficients w_k ($1 \leq k \leq K$).

$$S' = S_0 + \sum_{k=1}^K w_k (S_k - S_0) \quad (1)$$

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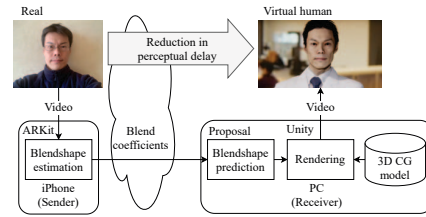


Figure 1: Overview of the proposed system

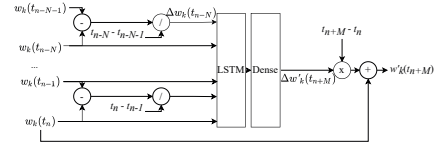


Figure 2: The proposed blendshape prediction

where the value range of the blend coefficients is from 0 to 1. To our knowledge, there is no conventional method that predicts blendshapes with LSTM, and the proposed method is the first attempt to predict future blendshapes from past blendshapes.

3 PROPOSED METHOD

Fig. 1 provides an overview of the proposed system. To render each other's virtual human, the inverted system is added to Fig. 1. A 3D CG model of a virtual human that supports blendshapes has been prepared in advance on the PC. The blend coefficients of the user's facial expressions are estimated by ARKit [1] on the iPhone. The blend coefficients are transmitted to the PC in real-time, and the proposed method predicts the future blend coefficients. The PC renders a virtual human with the predicted blendshape. Fig. 2 shows the proposed blendshape prediction method. The proposed method uses an LSTM to train the time-series variation of blend coefficients. Since the time needed for ARKit to estimate the blend coefficients fluctuates due to the effects of background processing, it may have a negative impact on conventional prediction methods such as LSTM. The proposed method trains the blend coefficients including their derivative values to reduce the effect of fluctuations. The derivative $\Delta w_k(t_n)$ for the k -th blend coefficient $w_k(t_n)$ at time t_n is obtained by Eq. 2

$$\Delta w_k(t_n) = \frac{w_k(t_n) - w_k(t_{n-1})}{t_n - t_{n-1}} \quad (2)$$

A set of K blend coefficients at a certain time is defined as one sample. The input and output of the proposed method are the blend coefficients w_i and derivative values Δw_i of the past $N = 16$ samples respectively, and the method is trained to predict the blend coefficients and derivative values of M samples ahead.

In the inference process, the predicted derivative is multiplied by the elapsed time and added to the most recent blend coefficient.

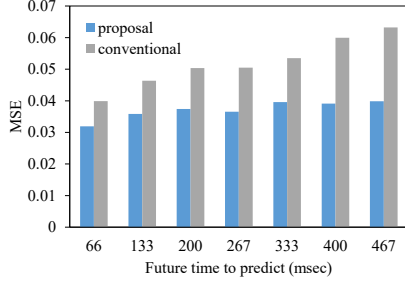


Figure 3: Comparison of MSEs for predicting the future at each time

Specifically, the blend coefficients $w'_k(t_{n+M})$ at the future time t_{n+M} can be obtained from the inferred derivative $\Delta w'_k(t_n)$ and the blend coefficient $w_k(t_n)$ used in the inference by Eq. 3.

$$w'_k(t_{n+M}) = w_k(t_n) + \Delta w'_k(t_n) * (t_{n+M} - t_n) \quad (3)$$

The proposed method absorbs the fluctuations in processing and transmission times by estimating the derivative and applying the elapsed time.

4 EXPERIMENTAL RESULTS

The accuracy of facial expression prediction was compared and evaluated on an iPhone 12 Pro and a notebook PC (CPU: Intel Core i7-10870H @ 2.20GHz, GPU: NVIDIA GeForce RTX 3070). For the virtual human, we prepared a 3D CG model that supports ARKit blendshapes ($K = 52$). Assuming a use case that involves a dialogue with a virtual human, the dataset recorded 54 minutes of blend coefficients uttered by seven people. They were asked to read out phonemically balanced sentences [7] for the various utterances.

The future $M = (2|4|6|8|10|12|14)$ samples were used as ground truth. To compare performance, the LSTM [2] was used as the conventional method. The number of LSTM units was set to 256 as in the conventional method [2]. The conventional method only inputs past $K = 52$ blend coefficients and directly predicts future blend coefficients. Unlike the conventional method, the proposed method inputs 104 (52 blend coefficients + 52 derivative values). Adam was used for optimization, and mean-squared-error (MSE) of the blend coefficients was used for the loss function.

Fig. 3 shows the comparison of MSE in the proposed method and the conventional method. The longer the prediction time is, the larger the MSE of the conventional method becomes. The proposed method has a relatively small increase in the MSE. In a preliminary experiment, the delay was about 250 msec with a Wi-Fi connection. Fig. 4 shows the 267 msec future prediction of the blend coefficient for the right eye. The conventional method significantly differs from the ground truth, being generally flat and not able to predict large changes. On the other hand, the proposed method has a relatively small difference from the ground truth. The other blend coefficients showed the same trend. Fig. 5 shows an example of applying the predicted blend coefficients to a virtual human. The MSE of the proposed method is about 0.037, but we can see that the proposed method is able to predict almost the same expression as the ground truth. The supplementary video shows the accuracy of the proposed method and the mismatching problem of the conventional method.

The average inference time is 34.8 msec / sample, so facial animation can be performed at about 15 frames per second when considering about 30 msec for transmission and rendering. The proposed method uses a simple LSTM for high-speed processing and improves accuracy by using derivative correction.

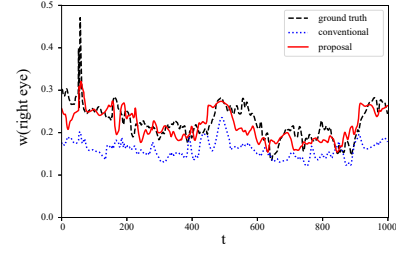


Figure 4: predicted blend coefficient. black-line: the ground truth (267 msec into the future), blue-line: result predicted by the conventional method. red-line: result predicted by the proposed method



Figure 5: Blendshape prediction comparison. left-top: the ground truth (267 msec into the future from the input image), right-top: the image input to the two methods. left-bottom: result predicted by the conventional method. right-bottom: result predicted by the proposed method

5 CONCLUSION

In this paper, we have proposed a framework to reduce the delay in rendering virtual humans by using facial expression prediction. The proposed method assumes temporal variation in blendshape estimation and trains the derivative of the blendshape by LSTM. We confirmed that the proposed method can adapt to temporal variation and reduce the error by 27% compared to the conventional method.

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